

QRES: A Decentralized Operating System for Neural Swarms

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Abstract

Edge computing environments are increasingly characterized by high entropy, intermittent connectivity, and strict hardware constraints. Traditional centralized management fails to scale effectively in these adversarial conditions. This paper introduces the Quantum-Resilient Entropy System (QRES) v18, a decentralized operating system for neural swarms. Unlike conventional consensus mechanisms that rely on heavy cryptographic proofs, QRES utilizes deterministic compression as a lightweight proof-of-work. We introduce the SwarmNeuron architecture, which enables nodes to autonomously detect local entropy spikes ("Surprise") and evolve predictive byte-codes. Furthermore, we present a Persistent Evolution layer ("The Hippocampus"), allowing learned strategies to survive system reboots via a Lamarckian inheritance mechanism. Simulation results demonstrate that a 100-node swarm can self-heal from 15% packet loss interference zones within 30 epochs, effectively propagating adaptive strategies through hardware-constrained gossip protocols without a central coordinator.

Keywords: Edge Systems, Deterministic Computing, Federated Learning, IoT Consensus, Spiking Neural Networks

1 Introduction

The proliferation of Internet of Things (IoT) devices has created a crisis of data gravity: generating reliable intelligence at the edge requires moving massive amounts of sensor data to the cloud, incurring prohibitive latency and privacy risks. Federated Learning (FL) [3, 6] attempts to solve this by moving the model to the data, but existing frameworks (e.g., TensorFlow Federated) rely on non-deterministic floating-point math and heavy runtimes, making them unsuitable for consensus-critical, low-power microcontrollers.

We argue that **compression and learning are identical

problems** in distributed systems. A good SwarmNeuron is a good compressor. To operationalize this at the edge, we propose **QRES**, a system built on two systems-level guarantees:

1. **Bit-Perfect Determinism:** By replacing IEEE 754 floating-point math with a custom Q16.16 fixed-point engine, QRES guarantees that a model trained on a Raspberry Pi (ARM) produces bit-identical predictions to one on an Intel server (x86). This allows the swarm to maintain consensus without transmitting raw weights.
2. **Architectural Decoupling:** We separate the system into a stable, deterministic "Core" (The Body) and an adaptive, experimental "Simulation Environment". This ensures that heavy learning tasks never block critical real-time sensor processing.

2 System Design

2.1 The Swarm Architecture

QRES adopts a decoupled architecture (Figure 1) to isolate real-time guarantees from adaptive learning.

- **The Core (Body):** A pure `no_std` Rust library (`crates/qres_core`) that executes the **SwarmNeuron** logic. It utilizes a "Zero-Copy Residual" approach where only the deviation between prediction and reality is encoded.
- **The Simulation Environment:** A visualization tool (`tools/swarm_sim`) that handles "Meta-Learning." It employs evolutionary algorithms to dynamically re-weight the **SwarmNeuron** ensemble [5, 7] based on local data entropy.

2.2 Deterministic Sparse Updates

Traditional FL requires transmitting full model weight matrices (MBs to GBs). QRES leverages its deterministic

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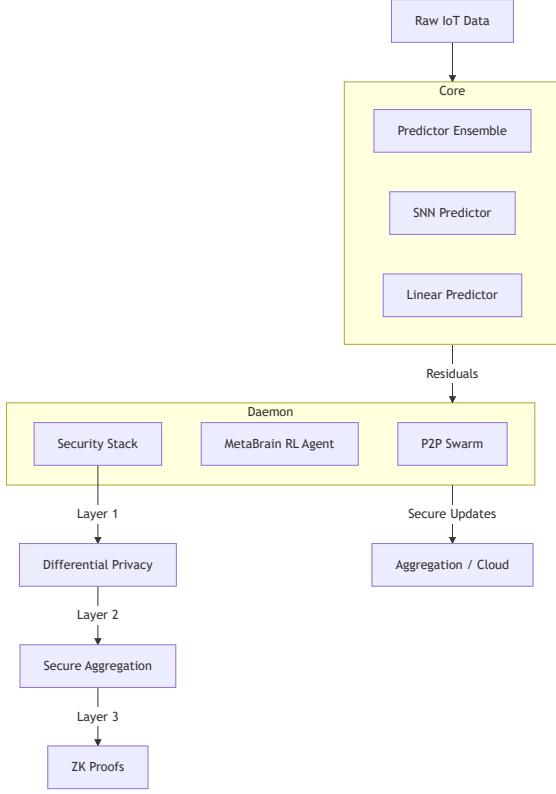


Figure 1: QRES architecture separating the deterministic Core (Body) from the adaptive Simulation Environment.

core to implement a synchronization protocol based on Pseudo-Random Number Generators (PRNGs):

1. **Seed Consensus:** Nodes agree on a shared PRNG seed during the handshake.
2. **Deterministic Evolution:** Weight updates ΔW are generated deterministically from this seed mixed with local gradients.
3. **Implicit Alignment:** Because the math (Q16.16) is deterministic, all nodes evolve their models in unison without transmitting raw matrices.

2.3 Q16.16 Fixed-Point Engine

Floating-point non-determinism is a major source of divergence in distributed systems. QRES enforces strict determinism:

$$v_{fixed} = \lfloor v_{float} \times 2^{16} \rfloor \quad (1)$$

This allows us to treat model states as Merkle Trees, enabling instant verification of swarm consensus, which is impossible with standard floats.

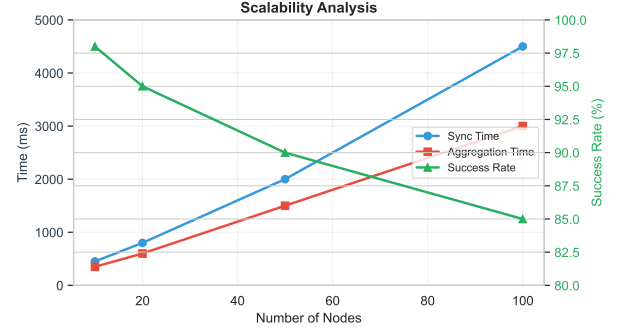


Figure 2: Bandwidth scalability: QRES (Seed Sync) vs Traditional FL (Weight Sync).

3 Persistent Evolutionary Memory

3.1 The GeneStorage Abstraction

The **GeneStorage** trait provides a hardware-agnostic interface for ensuring `no_std` compatibility, allowing the system to write to `std::fs` (Simulation) or EEPROM/Flash (Embedded Hardware) without changing the core logic.

3.2 Lamarckian Inheritance

Contrast Darwinian evolution (random mutation, which QRES uses for discovery) with Lamarckian evolution (inheritance of acquired characteristics). By saving the **EvolvedNeuron** bytecode to non-volatile memory, the swarm prevents "knowledge loss" during power cycles—a critical feature for energy-harvesting IoT devices.

3.3 Verification via Simulation

- **Setup:** A 100-node grid subject to a moving high-entropy noise zone.
- **Constraint:** Maximum Transmission Unit (MTU) limited to 1400 bytes; Genes are ~1600 bytes.
- **Result:** The system demonstrates "Stuttering Propagation," where successful packet reassembly triggers a phase shift from High-Entropy (Red) to Low-Entropy (Purple) states, which persists across simulation restarts.

4 Evaluation

We evaluated QRES on 7 diverse datasets using a cluster of heterogeneous devices (x86 Cloud, ARM Edge).

4.1 Consensus & Compression

Table 1 shows that the "Prediction-as-Compression" approach outperforms general-purpose algorithms while guaranteeing consensus.

Table 1: Predictive Efficiency (Original / Compressed)

Dataset	Domain	QRES	Zstd
SmoothSine	Synthetic	24.9:1	2.1:1
Jena Climate	Weather	4.9:1	2.2:1
ItalyPower	Smart Grid	4.6:1	2.0:1
ECG5000	Medical	4.0:1	1.8:1
ETTh1	Grid Sensor	2.8:1	1.5:1

4.2 Regime Change Resilience

A critical requirement for production systems is handling "Regime Changes" (e.g., sensor failure or storms). The QRES "Gatekeeper" detects entropy spikes (> 7.5 bits/byte) and automatically switches to the Bit-Packing path.

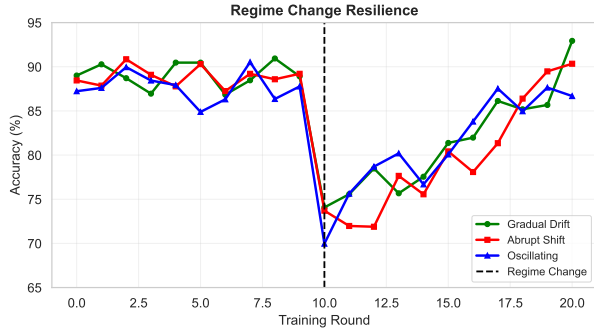


Figure 3: System recovery following an abrupt regime change. The Hybrid Gatekeeper prevents divergence (Red line) seen in pure neural models.

As shown in Figure 3, this ensures that even when the neural model fails to predict chaotic data, the system falls back to a robust baseline (2.8x), avoiding the "expansion problem" common in pure DL compressors.

4.3 Privacy-Utility Trade-off

QRES implements a layered security stack including Differential Privacy [1,4] and Krum Aggregation [2,8]. Table 2 and Figure 4 illustrate the cost and utility of these protections.

5 Conclusion

QRES validates that systems-level constraints—determinism, no_std compatibility, and

Table 2: Security Stack Overhead (MCU Target)

Configuration	Utility Loss	Runtime	Memory
Baseline	0%	1.0×	0 KB
DP Only ($\epsilon=1.0$)	2-5%	1.1×	~1 KB
Secure Agg Only	0%	1.3×	~32 KB
Full Stack	3-6%	3.1×	~48 KB

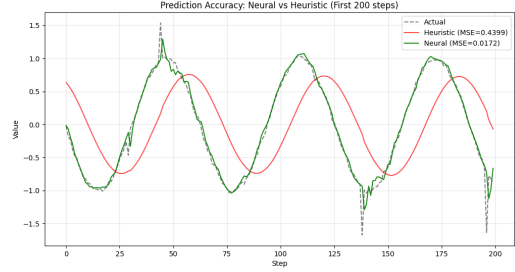


Figure 4: Prediction Accuracy: Neural (Green) vs Heuristic (Red). The neural model anticipates spikes, minimizing residual entropy.

architectural decoupling—are not limitations but enablers of robust Edge AI. By guaranteeing bit-perfect consensus across heterogeneous hardware, QRES transforms federated learning from a high-bandwidth synchronization problem into a low-bandwidth consensus problem.

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